

Ex. no: 2a)

Date: 29/01/25

Shell Script

Aim:

To write a Shellscrip to to display basic calculator.

Program:

```
#!/bin/bash

read -p "Enter first number: " a
read -p "Enter second number: " b

echo "Choose operation: + - * / %"
read op

case $op in
  +) echo "Result: $((a + b))" ;;
  -) echo "Result: $((a - b))" ;;
  \*) echo "Result: $((a * b))" ;; # escape * with \
  /)
    if [ $b -eq 0 ]; then
      echo "Division by zero not allowed"
    else
      echo "Result: $((a / b))"
    fi
  ;;
  %) echo "Result: $((a % b))" ;;
  *) echo "Invalid operation" ;;
esac
```

Sample Input and Output

Run the program using the below command

```
[REC@local host~]$ sh arith.sh
```

Enter two no

5

10

add 15

sub -5

mul 50

div 0

mod 5c"

Result:

Thus, the program is executed successfully.

Ex. no: 2b)

Date: 29/01/25

Shell Script

Aim:

To write a Shellscrip to test given year is leap or not using conditional statement

Program:

```
#!/bin/bash
```

```
read -p "Enter a year: " year
```

```
if (( year % 400 == 0 )); then
```

```
    echo "$year is a leap year"
```

```
elif (( year % 100 == 0 )); then
```

```
    echo "$year is not a leap year"
```

```
elif (( year % 4 == 0 )); then
```

```
    echo "$year is a leap year"
```

```
else
```

```
    echo "$year is not a leap year"
```

```
fi
```

Sample Input and Output

Run the program using the below command

```
[REC @ local host~]$
```

```
sh leap.sh enter number
```

```
12
```

```
leap year
```

Result:

Thus, the program is executed successfully